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ARTICLE

Preimplantation genetic testing for mitochondrial DNA mutation: ovarian response to stimulation, outcomes and follow-up



BIOGRAPHY

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KEY MESSAGE

The presence of mutant mitochondrial DNA copies does not affect oocyte ability to respond to FSH or to complete oocyte maturation. This study will assist in counselling couples on the available possibilities, risks and expectations of reproductive success of IVF with PGT-M compared with natural conception and subsequent prenatal diagnosis.

ABSTRACT

Research question: How do carriers of pathogenic mitochondrial DNA (mtDNA) respond to ovarian stimulation?

Design: A single-centre, retrospective study conducted between January 2006 and July 2021 in France. Ovarian reserve markers and ovarian stimulation cycle outcomes were compared for couples undergoing preimplantation genetic testing (PGT) for maternally inherited mtDNA disease ($n = 18$) (mtDNA-PGT group) with a matched-control group of patients undergoing PGT for male indications ($n = 96$). The PGT outcomes for the mtDNA-PGT group and the follow-up of these patients in case of unsuccessful PGT was also reported.

Results: For carriers of pathogenic mtDNA, parameters of ovarian response to FSH and ovarian stimulation cycle outcomes were not different from those of matched-control ovarian stimulation cycles. The carriers of pathogenic mtDNA needed a longer ovarian stimulation and higher dose of gonadotrophins. Three patients (16.7%) obtained a live birth after the PGT process, and eight patients (44.4%) achieved parenthood through alternative methods: oocyte donation ($n = 4$), natural conception with prenatal diagnosis ($n = 2$) and adoption ($n = 2$).

Conclusion: To the best of our knowledge, this is the first study of women carrying a mtDNA variant who have undergone a PGT for monogenic (single gene defects) procedure. It is one of the possible options to obtain a healthy baby without observing an impairment in ovarian response to stimulation.

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INTRODUCTION

Mitochondria are semi-autonomous cytoplasmic organelles that generate adenosine triphosphate (ATP) by oxidative phosphorylation through the respiratory chain. Dysfunctional mitochondria can induce a group of heterogeneous genetic and clinically severe disorders called mitochondrial diseases. Mitochondria have their own DNA, the mitochondrial DNA (mtDNA), which codes for 13 respiratory chain proteins, two ribosomal RNAs and 22 transfer RNAs. Nuclear genes encode all other respiratory chain proteins. Consequently, all the modes of transmission are possible in mitochondrial diseases, autosomal recessive or dominant, X-linked, maternal and many sporadic cases.

Mitochondrial disease can be caused by a variant in the nuclear genes. The large number of nuclear genes involved in mitochondrial functions and the clinical heterogeneity of mitochondrial diseases make these variants relatively difficult to identify.

Variants in mtDNA are maternally inherited in humans (*Stewart and Chinnery, 2015*), as sperm mitochondrial proteins are ubiquitinated and degraded by proteasomes and autophagy following their entry into the oocyte (*Schon et al., 2012*). The main mutations and deletions of mtDNA are responsible for devastating symptoms grouped as syndromes such as MELAS (mitochondrial encephalomyopathy; lactic acidosis; stroke), MERRF (myoclonic epilepsy with ragged red fibres), and NARP (neuropathy, ataxia and retinitis pigmentosa). Mutations of MtDNA are usually heteroplasmic with a coexistence of wild-type and mutated molecules in the same cell or tissue. The clinical severity of these diseases varies between individuals according to the level of mutant load (% of mutant mtDNA molecules versus total number of mtDNA copies), also known as the heteroplasmy rate (*Gorman et al., 2016*).

Currently, no treatment is available for mitochondrial diseases. The treatment is essentially symptomatic and does not significantly change the evolution of the pathology. Genetic counselling on the risk of disease transmission to the offspring and available measures to prevent the birth of an affected child should be proposed to all women carrying mtDNA variants before they attempt to conceive. Counselling, however, is difficult as mitochondrial DNA diseases do not follow Mendelian inheritance. Although some couples choose to remain childless, use egg donation or adopt, others choose to use medical strategies to avoid the birth of an affected child, such as prenatal diagnosis (PND) or preimplantation genetic testing (PGT). Often, PND with a proposed medical termination of pregnancy is experienced as an emotional burden. An alternative is PGT after an IVF procedure, which provides the opportunity to achieve a pregnancy with a healthy baby without the psychological stress induced by the risk of terminating the pregnancy (*Platteau et al., 2002*).The objective of PGT for a mtDNA variant is to determine the mutant load of embryos to transfer the one with the lowest heteroplasmy rate. The mutant load cut-off to consider an embryo as healthy (threshold) was different for each couple. The threshold is determined by several associated factors, such as mother and ascendant symptoms associated with mutant load, familial and pregnancy history and published scientific data. In France, of the five PGT centres authorized for PGT activities, only one is experienced in PGT-M for mtDNA variants (Antoine Beclere Hospital/Necker-Enfants Malades Hospital, APHP). However, PGT remains challenging because of persistent questions about the capacity of carriers of pathogenic mtDNA to respond to ovarian stimulation, knowing that mitochondria play an essential role in oocyte maturation and functions as the human oocyte contains approximately 100,000 mitochondria (*Babayev and Seli, 2015*).

The present study includes the largest published series on carriers of pathogenic mtDNA in a specialized department and explores the scope and results of PGT for women carrying a mtDNA variant. Our objectives were to compare the following: the response to ovarian stimulation performed for PGT between carriers of pathogenic mtDNA who underwent PGT (mtDNA-PGT) and a control group (women with a wild karyotype who underwent PGT for male indications); to describe the PGT outcomes in this population; and to provide a follow-up of our patients regardless of the outcome of the PGT process (live birth or unsuccessful PGT).

MATERIALS AND METHODS

The present study is a monocentric retrospective observational study carried out between January 2006 and July 2021, and is associated with a case-control study (ovarian stimulation cycles of women with a normal karyoptype and who underwent PGT for male indications).

Patients

Mitochondrial DNA preimplantation genetic testing group

All women included in this study were heteroplasmic for a variant in mtDNA. Before inclusion in our PGT programme, women symptomatic for mtDNA-mutation underwent a clinical examination to ensure they have no contraindication for IVF and pregnancy. Moreover, all women were tested for ovarian reserve quantified by blood anti-Müllerian hormone (AMH) level and antral follicle count (AFC) determined by ultrasonography.

Control group

Each ovarian stimulation cycle of the mtDNA-PGT group was matched with three control ovarian stimulation cycles (ratio 1:3) with the context of PGT for male indications. Matching criteria were female age at stimulation ± 3 years; same ovarian stimulation protocol; starting FSH dose ± 50 IU; AFC on the first day of ovarian stimulation ± 5; and year of starting PGT procedure ± 3.

Ovarian stimulation protocols

The ovarian stimulation protocol and initial dose of gonadotrophins were based on patient age, AFC, AMH level, body mass index and previous ovarian stimulation. Ovarian stimulation was achieved using a long agonist or antagonist protocol as previously described (*Mayeur et al., 2020*). For the long agonist protocol, patients received a single dose of the gonadotrophin releasing hormone (GnRH) agonist triptorelin (3 mg intramuscular Decapeptyl SR) (Ipsen, Boulogne-Billancourt, France) on the first or 20th day of the menstrual cycle. Three weeks later, pituitary desensitization was confirmed by the detection of low serum levels of progesterone, oestradiol, LH and follicle size less than 10 mm. Then, recombinant FSH (follitropin alpha Gonal F®) [Merck Serono, Geneva, Switzerland]; Bemfola® [Gedeon Richter, France], follitropin beta Puregon® [MSD Organon, Oss, the Netherlands] or urinary FSH [menotropin—Menopur®] [Ferring Pharmaceuticals, Copenhagen, Denmark]) was started for a minimum of 6 days. Patients undergoing GnRH antagonist protocols received pre-treatment with oral contraceptive pill or oestradiol valerate

(Provames®). Daily subcutaneous injections of recombinant FSH or urinary FSH were started either 24 h after oestradiol cessation or 5 days after pill cessation and were administered for at least 5 days using the initial dose. From the sixth day of recombinant FSH treatment onward, daily FSH doses were adjusted according to oestradiol levels, the number of growing follicles, or both. Starting on day 6, a 0.25 mg GnRH antagonist (ganirelix [Orgalutran®] [MSD, Puteaux, France] or cetrorelix [Cetrotide®] [Merck-Serono, Lyon, France]) was administered daily. Ovarian stimulation was monitored by transvaginal ultrasound and blood assessment of oestradiol, progesterone and LH levels from day 6 or 8 of stimulation until HCG administration. Ovulation was triggered with HCG (Ovitrelle® 250 μg) (Serono, Aubonne, Switzerland) or triptorelin 0.2 mg (Decapeptyl®) (Ipsen, Boulogne-Billancourt, France) subcutaneous injection administered when at least three preovulatory follicles (16–22 mm in diameter) were observed. Oocytes were retrieved 36 h after HCG administration by transvaginal ultrasound-guided aspiration.

Preimplantation genetic testing procedure

Laboratory techniques

For the PGT procedure, retrieved cumulus oocyte complexes were cultured in 3 ml of IVF culture medium (Origio, Lyon, France) until denudation. All matured oocytes obtained were inseminated by intracytoplasmic sperm injection. Fertilization was checked 18 h after sperm injection, and zygotes were cultured in microdrops (40 μl) of SAGE 1-Step™ (Origio, Lyon, France) at 37°C under an ambient atmosphere enriched in 5% CO_2.

Embryo biopsy was scheduled on day 3 to allow fresh embryo transfer on day 5 (*Bacus et al. 2021*). Embryo biopsy was carried out in SAGE 1-Step™ (Origio, Lyon, France) after being placed for a few minutes in CA/Mg-free medium (Biopsy Medium™) (Origio, Lyon, France). A laser procedure was used for the biopsy (Fertilase®) (MTGMedical Technology Vertriebs-GmbH, Altdorf, Germany). Embryo biopsy was carried out for all embryos with at least 6 blastomeres, and less than 25% fragmentation. One to two blastomeres were removed (one cell in embryo with six blastomeres). Blastomere (s) were removed and rinsed in a drop of phosphate buffered saline supplemented with 0.1% polyvinyl alcohol (Sigma Aldrich, Saint-Quentin-Fallavier, France) before being deposited in 3 μl of lysis buffer under visual control. For each sample, a small volume of rinsing medium was similarly transferred and used as a negative template.

Genetic analysis

Blastomere mutant loads were assessed via semi-quantitative fluorescent polymerase chain reaction (PCR) tests with restriction enzyme digestion. These methods are much more dated than others, such as next-generation sequencing technologies, but they are highly sensitive and accurate, fast and require a small amount of starting DNA. Primers were chosen using oligo 6.0 software (Medprobe, Oslo, Norway). Cross-hybridization of oligonucleotide primers to genomic DNA was ruled out by PCR amplification on mtDNA-less *Rho0* cells (*Parfait et al., 1998*). A limited number of PCR cycles were conducted to minimize the risk of underestimating mutant load resulting from heteroduplex formation during the reaction. After digestion, PCR products (1 μl) were electrophoresed on an ABI 3130xl 24-capillary sequencer (Life Technologies, Carlsbad, CA, USA). The results were analysed with GeneMapper software (Life Technologies, Carlsbad, CA, USA). The mutant load was calculated by dividing the mutant peak area by the sum of the wild-type and mutant peak areas. For all mtDNA variants, the estimated mutant load detection threshold was 2%. The estimated inter-assay variation was also 2%.

Embryo transfer and luteal support

Up to two healthy embryos were transferred on day 5 after intracytoplasmic sperm injection under ultrasound guidance (Voluson 730 Expert) (General Electric Medical Systems, Paris, France) using a classic Frydman catheter® (CCD Laboratories, Paris, France).

The mutant load cut-off to consider an embryo as healthy was different for each couple. It was determined by several associated factors, such as mother and ascendant symptoms associated with mutant load, familial and pregnancy history and published scientific data. Supernumerary healthy embryos were cryopreserved if they reached the blastocyst stage on day 5 or day 6. Slow cooling method was used before 2013, and then vitrification method was used for cryopreserved blastocysts after 2013. No supernumerary embryos, however, were frozen before 2013. Blastocysts were vitrified using RapidVit Blast® kit (Vitrolife®, Vastra Frolunda, Sweden). The entire procedure was carried out at 37°C according to the manufacturer's instructions (cryoprotectant concentrations are not detailed in the commercial kits). Straws (RapidI) (Vitrolife®, Vastra Frolunda, Sweden) with one blastocyst loaded were then immediately sealed and immersed in liquid nitrogen before storage.

For fresh transfer, the luteal phase was supported with natural micronized progesterone 600 mg/day and oral oestrogen supplementation (4 mg/day), starting the evening of embryo transfer (Utrogestan® 200 mg [Besins Heathcare] and Provames® 2 mg [SERB, Paris, France) and until the pregnancy test. Treatment was continued for up to 8 weeks of amenorrhoea in cases of pregnancy.

For cryopreserved embryo transfers, the endometrium was prepared with a programmed cycle regimen using oestrogen priming at a dose of 4 mg daily (Provames® 2 mg) (SERB, Paris, France) on day 1–3 of the menstrual cycle. Progesterone treatment (micronized progesterone vaginal [Progestan® 800mg] [Besins, Paris, France]) was added when endometrial thickness reached 7 mm or more. Clinical pregnancy was defined as the presence of one or more gestational sacs on ultrasound scans carried out at about 7 weeks of amenorrhoea.

Follow-up

Data were collected by telephone interviews to obtain information about the alternatives chosen by couples after their PGT procedure. Eleven questions were used to assess the following: the occurrence of pregnancy after stopping the PGT procedure at our centre; if patients were still in a relationship with the same partner; how they experienced emotionally and physically the PGT process; if they needed psychological support; if couples were still being treated in our centre; if they considered oocyte donation; if they considered adoption; if they had further infertility treatment in other centres; if they had considered natural conception; if they had undergone a prenatal molecular testing; and if the decision to stop the IVF procedure was burdensome.

Follicular output rate calculation

Follicular output rate (FORT) was used to assess antral follicle responsiveness to

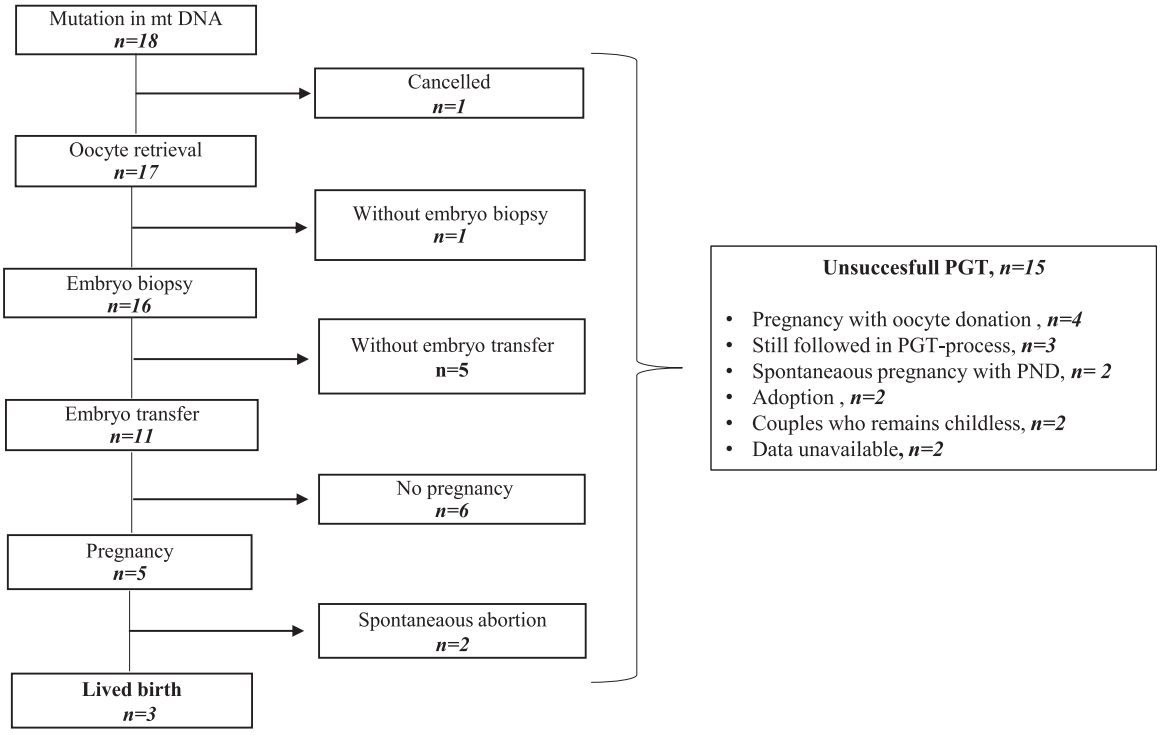


FIGURE 1 Outcomes of preimplantation genetic testing (PGT) in women with a mitochondrial DNA mutation. PND, prenatal diagnosis.

recombinant FSH independently of the follicular cohort size before treatment, as previously described (*Genro et al., 2011*). Ovarian ultrasound scans were systematically carried out before each ovarian stimulation using a 5.0–9.0 MHz multifrequency transvaginal probe (Voluson 730 ExpertR) (General Electric Medical Systems, Paris, France). The objective of ultrasound examination was to evaluate the number and sizes of small antral follicles. All follicles measuring 3–22 mm in mean diameter (mean of two orthogonal diameters) in each ovary were counted. Only antral follicles less than 8 mm were considered for AFC at baseline (d0), and follicles between 16 and 22 mm on day of ovulation triggering were considered for preovulatory follicle count. To optimize the reliability of ovarian follicular assessment, the ultrasound scanner was equipped with a tissue harmonic imaging system, which allowed improved image resolution and adequate recognition of follicular borders. The FORT was calculated by the ratio between the preovulatory follicle count on the day of ovulation triggering and AFC at day 0 × 100. The choice of considering only 16–22 mm follicles for the calculation of FORT was validated in previous investigations (*Genro et al., 2011*; *Gallot et al, 2012*).

Statistical analysis

The results are presented as the mean ± SD or median and interquartile range (25–75%) for non-normal distributions. All statistical tests were two-sided, and $P < 0.05$ was considered statistically significant. All women with mtDNA variants and their ovarian stimulation cycles with oocyte retrieval were included in the statistical analysis. Ovarian stimulation outcomes were compared between the mtDNA-PGT group and matched controls (ratio 1:3) by univariate conditional logistic regression models. NCSS Statistical Software, 2021 (NCAA, LLC, Kaysville, Utah, USA) was used to analyse the data.

Ethical approval

Consent from all patient couples was obtained at the time of PGT for the use of their medical data for research. Our database was approved on 21 February 2007 by the National Data Protection Authority (Commission Nationale de l'Informatique et des Libertes, CNIL number 1217921). An Institutional Review Board (2021-6) approved this study on 25 November 2021.

RESULTS

Throughout the study period, the focus was on PGT for mtDNA variant for 18 women enrolled into the study and who underwent 32 ovarian stimulation cycles with oocyte retrieval (mtDNA-PGT group) (FIGURE 1).

Patient characteristics

The baseline characteristics of the 18 mtDNA-mutated women included in the present study are presented in TABLE 1. In the study group, 10 women carried the m.3243A>G variant in the *MT-TL1* gene responsible for MELAS syndrome, three carried the m8344A>G variant in the *MT-TK* gene responsible for MERRF syndrome, two carried the m.8993T>G/C variant in the *MT-ATP6* gene responsible for NARP syndrome, two carried the m.9185T>C variant in *MT-ATP6* gene responsible for NARP syndrome and three carried the m.10197G>A variant in the *MT-ND3* gene responsible for Leigh syndrome. Heteroplasmy levels were assessed for each 18 patients in blood samples and varying from 2–78%. Seven women carriers of pathogenic mtDNA were symptomatic, with symptoms varying from diabetes, hearing loss, epilepsy, cardiomyopathy depending on the syndrome and mutant load rate in blood samples. The other patients (except one for which data was not available) discovered their mtDNA variant through a family study or because of the birth of an affected child.

TABLE 1 BASELINE CHARACTERISTICS OF WOMEN WITH MITOCHONDRIAL DNA

Patient	Age, years	Cycles, n	Syndrome	Gene	Mutations	HP,%	Diagnostic	AFC	AMH, ng/ml	Embryos, n	Embryos biopsied, n[a]	Embryos transferred, n	Pregnancies, n	Live births, n
1	34	1	Leigh	MT-ATP6	m.10197G>A	70	Affected child	13	1.7	1	1 (0)	0	0	0
2	33	1	Leigh	MT-ATP6	m.10197G>A	25	Affected child	20	4.4	7	1 (1)	1	0	0
3	38	1	MELAS	MT-TL1	m.3243A>G	30	Symptoms	23	1.55	7	4 (3)	1	0	0
4	31	3	MERRF	MT-TK	m.8344A>G	50	Symptoms	14	1.4	15	10 (3)	0	0	0
5	35	1	MELAS	MT-TL1	m.3243A>G	36	Symptoms	11	1.2	4	2 (1)	1	0	0
6	35	1	MELAS	MT-TL1	m.3243A>G	25	Symptoms	13	1.4	1	1 (0)	0	0	0
7	27	1	MERRF	MT-TK	m.8344A>G	75	Affected child	10	3.6	14	9 (3)	2	1	1
8	35	1	Leigh	MT-ATP6	m.10197G>A	36	Affected child	24	3.54	3	3 (1)	1	0	0
9	35	4	MELAS	MT-TL1	m.3243A>G	7	Family study	21	1.85	25	14 (12)	7	0	0
10	27	3	NARP	MT-ATP6	m.9185T>C	19	Family study	24	2.8	22	8 (4)	2	0	0
11	33	1	MELAS	MT-TL1	m.3243A>G	22	Symptoms	12	1.29	3	0	0	0	0
12	35	1	MELAS	MT-TL1	m.3243A>G	2	Family study	18	4.42	4	4 (4)	2	1	1
13	27	2	MELAS	MT-TL1	m.3243A>G	40	NA	10	1.1	3	2 (1)	0	0	0
14	37	3	MELAS	MT-TL1	m.3243A>G	22	Affected child	19	1.85	28	23 (10)	6	1	1
15	31	2	MELAS	MT-TL1	m.3243A>G	30	Symptoms	13	1.55	6	1 (1)	0	0	0
16	38	5	NARP	MT-ATP6	m.8993T>C	15	Family study	30	3.6	35	22 (6)	5	1	0
17	30	2[b]	MELAS	MT-TL1	M3243A>G	30	Symptoms	14	1.80	0	0	0	0	0
18	29	1	MERRF	MT-TK	M8344A>G	78	Family study	40	4.87	9	5 (2)	1	1	0
Total	33[c] [30–35]	1.8[d] (1–5)	/	/	/	34[d] (2–78)	/	16[c] [13–22.5]	1.8[c] [1.4–3.6]	11.5[d] (0–35)	EB:6.1 (0–23)[d] GT:2.9 (0–12)[c]	1.6[d] (0–6)	5 (27.8%)	3 (16.7%)

[a] Embryo considered transferable.

[b] Both cycles were cancelled owing to insufficient ovarian response.

[c] Expressed as median [25p–75p] or

[d] (minimum–maximum).

AFC, antral follicle count; AMH, anti-Müllerian hormone; HP, heteroplamy rate of female patient (assessed in patients' blood sample); MELAS, mitochondrial, encephalomyopathy, lactic acidosis, stroke; MERRF, myoclonic epilepsy with ragged red fibres; NA, not available; NARP, neuropathy, ataxia and retinitis pigmentosa.

TABLE 2 OVARIAN STIMULATION CYCLE OUTCOMES IN PREIMPLANTATION GENETIC TESTING AND CONTROL GROUPS

	Affected women (n = 17; 32 cycles)	Partners of affected men (n = 96; 96 cycles)	P-value
Age, years[a]	34 (30–37)	34 (31–37)	
AMH, ng/ml[b]	1.8 (1.4–3.6)	2.7 (1.7–3.6)	0.14
AFC[c]	16 (12.5–23.5)	17.5 (12–21)	
Maximal oestradiol, pmol/l	2658 (1606–3711)	2578 (1886–3168)	0.4
FORT, %	40. 8 (22.3–60.4)	40 (29.6–54.4)	0.26
Duration, days	12 (11–13)	11 (10–12)	0.02
Total gonadotrophin dose, IU	3149 ±1008	2767 ± 745	0.003
Retrieved oocytes, n	10.5 (6–13)	10 (7–14)	0.75
Mature oocytes, n	9 (5–11)	9 (6–13)	0.38
Protocol			0.44
Agonist, %	34.38	35.42	
Antagonist, %	65.63	64.58	
Cleaved embryos, n	4.5 (3–8)	6 (3–9)	0.33

Results expressed as mean ± SD or median (25th–75th percentile), unless stated otherwise.

Results expressed as mean ± SD or median (25th–75th percentile), unless stated otherwise.

[a] Age at beginning of first cycle.

[b] Age before the start of the first cycle in women who have undergone multiple cycles.

AFC, antral follicular count; AMH, anti-Müllerian hormone; FORT, follicular output rate.

Ovarian stimulation cycle outcomes in the group undergoing preimplantation genetic testing for maternally inherited mitochondrial DNA disease

The results of the 32 ovarian stimulation cycles leading to oocytes retrieval (n = 17) are presented in TABLE 2. The number of ovarian stimulation cycles per woman ranged from one to five (the median was one [one to three]). At the beginning of the cycles, the median age was 34 years (30–37) years, the median AFC was 16 (12.5–23.5) follicles, and the median AMH serum level was 1.8 (1.4–3.60) ng/ml. The median number of oocytes recovered per cycle was 10.5 (6.0–13.0), leading to a median of nine (five to 11) metaphase II oocytes per cycle.

Ovarian stimulation cycle outcomes in the group undergoing preimplantation genetic testing for maternally inherited mtDNA disease versus control group

The 32 ovarian stimulation cycles leading to oocyte retrieval carried out in women with mtDNA were matched with 96 ovarian stimulation cycles carried out in the context of PGT for male indications. Women in the mtDNA group needed a longer ovarian stimulation as the duration of the ovarian stimulation was significantly longer than in the control group (12 [11–13] versus 11 [10–12] days, P = 0.02). In line with this result, the total dose of gonadotrophins was significantly higher in ovarian stimulation for carriers of pathogenic mtDNA (3149 ± 1008 IU versus 2767 ± 745 IU in the control group) (P = 0.003). Other parameters of ovarian response to FSH (maximum oestradiol and FORT), as well as the number of retrieved and matured oocytes per cycle, were not significantly different between the two groups.

Preimplantation genetic testing outcomes in the group undergoing preimplantation genetic testing for maternally inherited mitochondrial DNA disease

The PGT outcomes are presented in FIGURE 1. Among the 18 patients, 17 underwent oocyte retrieval and one did not respond to ovarian stimulation. The number of cleaved embryos showed no significant difference between women with mtDNA mutation and partners of affected men (4.5 [3–8] versus 6 [3–9]; P = 0.33), as well as the number of biopsied embryos. Sixteen patients (29 cycles) had at least one embryo biopsied. Among them, 10 patients had at least one cycle with at least three embryos biopsied. In total 56.2% (18/32) of cycles in the mtDNA group had at least three embryos biopsied versus 61.5% of total cycles (59/96) in the control group (P = 0.60). Fourteen patients corresponding to 21 cycles (65.6%, 21/32) had at least one healthy embryo (two patients had no healthy embryos). Eleven patients corresponding to 18 cycles (56.3%, 18/32) underwent a fresh embryo transfer with a mean of 1.61 ± 2,17 embryo per transfer. Among these couples, four couples also had a supernumerary frozen embryo transfer with a mean of 1.12 ± 0.35 embryo per transfer.

Of the other three women with healthy embryos and no embryo transfer, two did not undergo embryo transfer because their embryo was of poor quality (one fragmented embryo, one with embryonic arrest) and for one woman no transfer was carried out because the couple was too anxious to transfer an embryo with a heteroplasmy rate of 30%.

A total of five pregnancies and three live births were obtained after fresh embryo transfer, and no live births were achieved after supernumerary frozen embryo transfer. Overall, the clinical pregnancy rate per embryo transferred was five out of 29 (17.2%) and live birth rate per embryo transferred was three out of 29 (10.3%).

Follow-up of patients after preimplantation genetic testing

Among the 18 patients, two of them did not answer our phone call (TABLE 3), and the remaining patients (n = 16) completed our survey. After PGT, most of the couples were still of childbearing age (33 ± 3 years) and were still in a relationship, even though 12 couples found their IVF process emotionally and physically difficult (TABLE 3). Only two of them, however, had access to psychological support. Three patients achieved their parental goal thanks to IVF with PGT-M. Three patients were still followed in our centre at the time of the study. Nine considered alternatives to IVF with PGT-M and achieved a pregnancy with oocyte donation (n = 4) or after natural conception with PND (n = 2). Three patients considered adoption, and two of them achieved parenthood through it, the other one failed. Unfortunately, two couples remained childless and decided to stop trying to have a child, one right after the IVF with PGT-M in our centre and the other one after trying other unsuccessful alternatives.

DISCUSSION

To the best of our knowledge, this is the first study investigating ovarian response to stimulation and PGT outcomes for patients

TABLE 3 PATIENTS' ANSWERS TO THE PREIMPLANTATION GENETIC TESTING FOLLOW-UP SURVEY

Survey questions	Yes, *n* (%)	No, *n* (%)
Did you achieve your parental project?	14 (87.5)	2 (12.5)
Are you still in a relationship with your partner?	16 (100)	0
Did you seek any psychological assistance?	2 (12.5)	14 (87.5)
Did you find your IVF process emotionally and physically difficult?	12 (75)	4 (25)
Are you still undergoing IVF in our centre?	3 (18.8)	13 (81.3)
Did you consider oocyte donation?	4 (25)	12 (75)
Did you consider adopting?	3 (18.8)	13 (81.3)
Did you consider trying IVF in another country?	3 (18.8)	13 (81.3)
Did you consider natural conception?	2 (12.5)	14 (87.5)
If yes, did you consider prenatal diagnosis?	2/2 (100)	0
Was the decision to stop IVF burdensome for you?	11 (68.8)	5 (31.3)

Sixteen patients answered our survey; two remained unavailable.

with pathogenic mtDNA variant. This report includes a total of 18 women carriers of pathogenic mtDNA who underwent a total of 32 ovarian stimulation cycles with oocyte retrieval. In this population, ovarian stimulation outcomes were not altered compared with matched ovarian stimulation cycles carried out in the context of PGT for male indications. Overall, three patients (16.7%) obtained a live birth after the PGT process, and eight patients (44.4%) achieved parenthood through alternative options.

The present study focuses on a very specific PGT indication and reports 15 years of experience. We are aware, however, that our group may not be representative of the wider population with pathogenic mtDNA variant.

Oocyte developmental competence is defined as the ability of an oocyte to resume meiosis and complete nuclear maturation, as well as to develop into embryos after fertilization and ultimately into live offspring (*Swain and Pool, 2008*). Studies conducted in many mammalian species have demonstrated that the quality of the cytoplasmic environment is crucial for sustaining oocyte developmental competence due to key mitochondrial roles (*Bao et al., 2003*; *Comizzoli et al., 2011*; *Nakagawa et al., 2011*). In humans, premigratory primordial germ cells have fewer than 10 mitochondria (*Van Blerkom, 2004*). Direct measurement of human mtDNA copy numbers estimates that the numbers of mitochondria within germ cells increase 10-fold when primordial germ cells reach the ovary (*Van Blerkom, 2004*) and continue to increase to approximately 100,000-fold in fully grown mammalian gametes (*Ramalho-Santos et al., 2009*). The increase in mitochondrial numbers also coincides with dynamic changes in the distribution and activity of mitochondria during folliculogenesis. This specific point has been evaluated in cat oocytes by confocal fluorescence microscopy using MitoTracker for different mature oocyte stages (*Songsasen et al., 2017*).

The primordial role of mitochondria in oocyte competence could also be demonstrated by ovarian ageing, with a deterioration of oocyte quality resulting from increased abnormalities of the nuclear genome and the deterioration of cytoplasmic quality(*Qiao et al., 2014*). Despite the indirect evidence of the involvement of mitochondria in oocyte quality, publications on this subject remain few, with small effects especially in humans (*Barritt et al., 1999*; *Shufaro et al., 2012*; *Boucret et al., 2015*; *May-Panloup et al., 2016*). The presence of mtDNA variant in an oocyte should result in a reduction in the absolute number of wild-type DNA copies. The influence of mtDNA variants on the ovarian response to stimulation has not yet been established. We published preliminary results in 2021, in which we found a potential effect of mtDNA variants on the ovarian reserve of patients, with AMH levels significantly lower in the group of patients with mtDNA variants than in the control group. We were not, however, able to draw any conclusions because the number of patients from the mtDNA group was limited ($n = 4$) (*Chatzovoulou et al., 2021*). In the present study, we could not confirm our previous hypothesis as we matched our patients on their ovarian reserve, based on the AFC, which is one of the main limitations of our study. We did not observe any impaired response to ovarian stimulation in mtDNA variant carrier women in terms of serum oestradiol level or number of retrieved or mature oocytes obtained compared with a matched control group. Furthermore, to quantify follicle responsiveness to exogenous FSH, we used FORT, a qualitative indicator of ovarian follicular competence (*Gallot et al., 2012*), in infertile patients. It has the advantage of being independent of AFC before FSH administration, allowing a better understanding of the capacity of follicles to respond to ovarian stimulation. We found no significant difference between the FORT of the mtDNA group and control group ($P = 0.26$). Nevertheless, we found that patients in the mtDNA group responded more slowly to ovarian stimulation, at a mean of 1 day longer than the control group, and in line with this, that they needed a higher total dose of gonadotrophins.

The presence of a high number of mtDNA copies in mature oocytes is also known to be mandatory for the successful early development of preimplantation embryos as they require a large amount of energy, as highlighted by several studies in various species (*Reynier et al., 2001*; *Spikings et al., 2007*; *Chiaratti et al., 2010*; *Wai et al., 2010*). As mtDNA content remains stable during early embryonic development (*Monnot et al., 2011*), the number of wild-type mtDNA copies is reduced for embryos bearing a mtDNA variant. This may induce early embryo development failure, protecting against transgenerational transmission of mutant mtDNA molecules (*Steffann et al., 2015*). Conversely, *Chatzovoulou* et al. *(2021)* demonstrated that embryo quality was not impaired by the presence of a maternal mitochondrial defect with no statistical difference of embryonic grades on day 3, day 4 or 5 between a control embryos group ($n = 165$ embryos) and a mitochondrial embryos group ($n = 16$ embryos). Authors stated that one of the limits of the study is the limited statistical power caused by the low number of available embryos ($n = 16$) with mtDNA variants (*Chatzovoulou et al., 2021*).

In the present study, the number of cleaved embryos as well as the number of biopsied embryos showed no significant

difference between women with mtDNA mutation and partners of affected men. Assessment of embryonic morphology was outside the scope of the present study, and, therefore, we can cautiously confirm that the mtDNA variant had no effect on early embryo development.

Unfortunately, the small number of mtDNA mutated patients (n = 18) included in the study did not allow us to statistically determine a cut-off value of mature oocytes for having at least one healthy embryo, and further investigation is required.

Two recent studies summarized and analysed how patients decide to use and experience PGT-M (*Pastore et al., 2019*; *Hughes et al., 2021*). Although PGT-M avoids medical termination of a naturally conceived pregnancy bearing an affected child, IVF with PGT-M for mtDNA variants can be a long and stressful experience. In our centre, couples must wait for more than 1 year before starting their PGT attempt. Moreover, the process of PGT could be extremely impractical and psychologically demanding. Indeed, 15 patients in this study had unsuccessful attempts and remained childless after the PGT process. Among them, seven had never had an embryo transfer. Although the embryo might be 'healthy' from a mtDNA mutation load perspective, it might not be a top or good-quality blastocyst, so the chance of pregnancy is low based on embryo quality. To have good pregnancy results it is necessary to have both a good-quality embryo and a genetic result compatible with a transfer. Patients must be informed of this challenge before making the decision to begin the PGT process. Unsuccessful PGT procedures can have long consequences for emotional wellbeing. Previously published survey studies of couples who had proceeded with IVF and PGT-M demonstrated that the psychological effects of the process remain present for up to 3 years after the attempt (*Järvholm et al., 2017*).

In our cohort study, among the 15 patients who had an unsuccessful PGT, eight considered alternatives to IVF with PGT and achieved a pregnancy through oocyte donation (n = 4) or after natural conception with PND (n = 2), and two patients adopted. Overall, most of our patients successfully concluded their parental project (three with PGT and eight with alternative methods). Most of the couples reported that PGT procedures were an important step in their parental project, whereby it was either successful or it helped them to move on to other options. No patient has informed us of the use of mitochondrial replacement therapy, such as maternal spindle or pronuclear transfer, whose use remains limited worldwide and is forbidden in France (*Greenfield et al., 2017*).

In conclusion the presence of mutant mtDNA copies do not affect oocyte ability to respond to FSH or to complete oocyte maturation This information provided here extends knowledge about the current state of ovarian stimulation for carriers of pathogenic mtDNA. In relation to the descriptive PGT-M outcomes described in this particular population and the retrospective follow-up conducted, the present study allows us to provide patients and medical personnel, personalized counselling on the available possibilities, risks and expectations of reproductive success of IVF with PGT-M compared with natural conception and subsequent PND.

DATA AVAILABILITY

Data will be made available on request.

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AUTHORS' ROLES

AM: study design, collection of data, interpretation of data, writing the manuscript, critical revision of the manuscript; EB: study design, collection of data, writing of the manuscript, interpretation of data; JB, CD, LH, KC, SM, MG: study design and critical revision of the manuscript; JS: collection of data, critical revision of the manuscript; NF: study design; analysis and interpretation of data; critical revisions of the manuscript; and CS: study design, collection of data, statistical analysis, interpretation of data, critical revision of the manuscript.

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